SOFTWARE REQUIREMENTS SPECIFICATION

for

ShadePaths

Version 1.1

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1. Introduction

This section of the Software Requirements Specification (SRS) document provides the purpose, scope, used terminology, list of references, and overview of the SRS document's structure.

1.1. Purpose

The purpose of this SRS document is to describe the requirements specifications for the ShadePaths software system. The document is primarily intended for review and approval by interested stakeholders and serves as a formal reference for the development team during the design and implementation of the initial version of the software system.

1.2. Scope

ShadePaths is a web mapping software system that provides shadow visualisations and navigation services. The system enables end-users to view shaded areas generated by buildings and calculate routes based on two routing priorities: shade coverage and estimated travel time.

The software system contains two component applications. The first component is an application that operates within the browser of the end-user's device, referred to as the ShadePaths Application (SPA). The second component is an application located on a web-server for querying a database containing geospatial datasets and weather datasets, referred to as the Geospatial Database Server (GDBS). The two components combined are referred to as "the system".

The system supports real-time location tracking and route generation. However, real-time traffic conditions and dynamic road events are out of scope.

1.3. Definitions, acronyms, and abbreviations

Term	Definition
Shall	verbal form to indicate that the requirement must be followed strictly
Should	verbal form to indicate that the requirement is optional, recommended to be followed, and not necessarily required
The system	The software system with two components: SPA and GDBS
Feature	A system or subsystem that is utilised to achieve a function
Requirement	A necessary condition that the system needs to operate as intended.
Stakeholder	Groups or organizations with an interest in utilizing the framework proposed through this project.
Users	The intended end-user of the system defined in this document.
Geospatial Data	Data that includes information related to locations on the Earth's surface.
Weather Data	Data that include meteorological information that describes atmospheric conditions at a given time and place.
External Data source	Other external systems with data outside the defined system.
Static	Contents that cannot be modified

Dynamic	Contents that can be managed and updated.
Stimulus	An event occurring in a system's environment that causes the system to react in some way
Response	A signal or message that a system sends to its environment
Subsystem	A system that is part of a larger system.
SPA	ShadePaths Application
GDBS	Geospatial Database Server
BMD	Base Map Displayer
SO	Shade Overlay
RS	Route Selector
NAV	Navigator
REQ	Required
OPT	Optional

Table 1: Terminology used in the SRS

1.4. References

[1] L. Gogorikidze, G. Svanadze, and M. Tsintsadze, "Shade-aware Routing For Sunburn Prevention," in *12th International Workshop on ADVANCEs in ICT Infrastructures and Services (ADVANCE 2025)*, Nice, France, Jun. 2025, pp. 30–37. DOI: `10.48545/advance2025-fullpapers-2\_1` [Online]. Available: `https://univ-evry.hal.science/hal-05308276`

[2] Y. Feng, P. Zhang, J. Xue, Z. Chen, and L. Meng, "Walking in the shade: Shadow-oriented navigation for pedestrians," in *Proceedings of the 32nd ACM International Conference on Advances in Geographic Information Systems*, ser. SIGSPATIAL '24, Atlanta, GA, USA: Association for Computing Machinery, 2024, pp. 677–680, ISBN: 9798400711077. DOI: `10.1145/3678717.3691287` [Online]. Available: `https://doi.org/10.1145/3678717.3691287`

[3] I. C. S. S. E. S. Committee, I. of Electrical, E. Engineers, and I.-S. S. Board, *IEEE Recommended Practice for Software Requirements Specifications* (IEEE Std). IEEE, 1998, ISBN: 9780738104485. [Online]. Available: `https://books.google.co.jp/books?id=MYBGAAAAYAAJ`

1.5. Overview

The remainder of this SRS contains an overall description of the system (Section 2) and the specific requirements for the system (Sections 3).

2. Overall Description

This section of the SRS describes the general factors that affect the system and its requirements. The information presented here provides context for understanding the detailed requirements specified later in this document (Section 3).

2.1. Product Perspective

The system is organized as a client-server web application consisting of the SPA and the GDBS. The SPA represents the client-side portion of the system, while the GDBS represents the server-side portion responsible for managing the geospatial database and calculating routes.

The system operates within a web browser environment and depends on external weather data services for normal operation. The relationship between the major system components and external entities is illustrated in Fig 1.

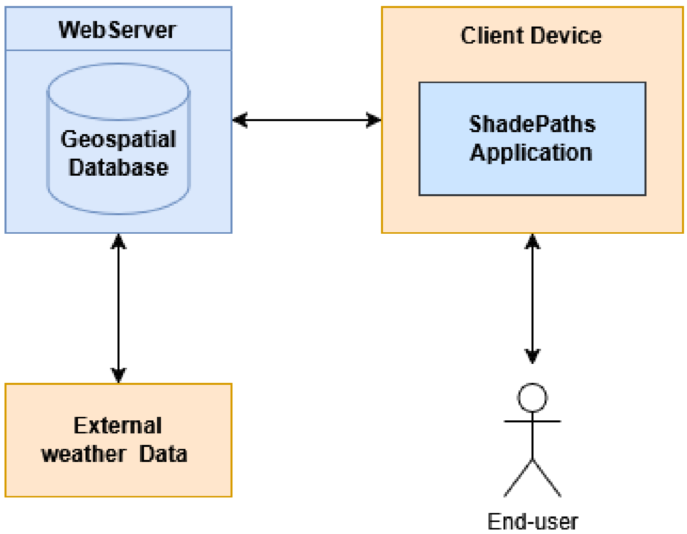

Figure 1: Block Diagram

2.2. Product Functions

The system provides an interactive map interface for displaying shade visualisations. It estimates shaded areas by processing building height, height above mean sea level, and solar position datasets. The calculation determines which areas are expected to be shaded at a given time. The system projects shaded areas for the current day and allows end-users to view projected shade coverage at different days and time through a slider interface. Additionally, it retrieves weather datasets and uses cloud cover data to estimate the predefined geographic region's shade coverage.

The system also calculates routes. End-users can enter a starting point and destination. The system generates possible walking routes with given user inputs. Multiple route options are then presented according to routing priorities such as travel distance and shade coverage. This allows the end-user to view and select a preferred route before navigation. During navigation, the system tracks and updates the user's current location.

2.3. User Characteristics

Intended users of the system are general pedestrians. Users are expected to possess general familiarity with modern web browsers, interactive maps and mobile device applications.Users are not expected to possess any specialised technical knowledge.

The system is intended for use by a broad range of users with varying levels of technical experience. User interface emphasises on ease of use and clear interface layouts.

2.4. Constraints

The system only supports routing and visualization within a predefined geographic region. The maximum possible route length is limited by the total supported geographic area. Only shadows generated by buildings and underground routes are considered during route calculations.

The system is designed for standard consumer devices with web browsing capabilities and may be affected by hardware limitations such as network availability, device performance, and location-service accuracy.

Except for weather datasets, all datasets managed by the GDBS remain static after initial deployment. Weather datasets are updated periodically to estimate shaded areas using cloud cover data.

2.5. Assumptions and Dependencies

All datasets managed by the GDBS and location data from end-user devices are assumed to be accurate. The solar position is assumed to remain consistent between equivalent dates each year for simplification.

Route calculations depend on the accuracy and availability of external weather data sources. If cloud cover across the predefined geographic region exceeds a defined threshold, the entire region is assumed to be covered in shadow.

It is assumed that internet connectivity is available during normal operation of the system. When a route is calculated from user location, the walking speed of the end-user is assumed to be the average walking speed of the general population.

3. Specific Requirements

This section of the SRS contains all of the software requirements in detail to enable designers to design a system that satisfies requirements, and allow testers to test that the system satisfies those requirements.
References to requirements throughout this SRS are structured accordingly to the following guidelines:

- <REQ >or <OPT >: REQ for requirements, OPT for optional recommendations.
- <Sublabel >: Short, two or three letter label for the high-level category of the entry.
- <Major number >: The major number is incremented for every major capability captured by a requirement.
- <Minor number >: The minor number starts with 1 for the first minor capability and is incremented by every entry that supports a major capability.

3.1. External Interface Requirements

This subsection provides a detailed description of all inputs into and outputs from the system.

3.1.1. User Interfaces

The SPA provides a web-based graphical user interface through a two-dimensional tiled map display. Multiple user interfaces allows users to interact with the map, view shade visualisations and access routing services.

The primary interface shall consist of a map viewing area, shade visualisation toggle and an option to enter navigation mode with "the route input" interface. Any error messages or notifications will be shown on the top most layer.
For Shade visualisations, there are two types of interfaces:

- The "ShadeButton" interface provides a toggle-able button that allow users to show and hide shade visualisations based on the current time.
- The "ShadeButton and Slider" interface contains all elements of the "ShadeButton" interface, and additionally provides a slider control input for the user when toggled on. Users can adjust the slider to view projected shade coverage at different times throughout a selected 24-hour period. The selected time value is displayed using a standard time representation.

To receive the starting location and destination directly from the user, the "User Input" interface provides an input overlay on top of the primary interface. A pin is shown in the center of the map and the user can enter the location by moving the map to match with the pin. The start location and destination are shown by a green and red pin respectively.

The "Route Selection" interface provides an overlay with calculated routes based on user input in the "User Input" interface. Multiple routes are shown in a distinct color along with its associated route information when selected. When initialised, none of the shown routes are selected, and the unselected routes show its main data (ETA for the shortest route, Shade percentage for the most shaded route, etc.). When a route is selected, all information of the route is shown in detail on the third half of the screen.

The "Navigation" interface displays the current user location, selected route, and estimated time to reach destination. The user will be represented by a blue triangle icon.

3.1.2. Hardware Interfaces

All client-side components shall support personal computers with modern browsers.The system shall use location data from the user's device, such as GPS or Wi-Fi.

3.1.3. Software Interfaces

The system shall interact with external data sources to receive cloud cover data.

3.1.4. Communication Interfaces

Data exchanged between the SPA and GDBS shall be transmitted through HTTP-based communication interfaces using structured data formats.

3.2. System Features

This subsection describes each internal subsystem and its components. This includes: a brief description, the introduction/purpose of the feature, the sequence of events following user interaction, and the associated functional requirements.

The subsystem names are summarized below along with their abbreviations:

- Base Map Displayer (BMD)
- Shade Overlay (SO)
- Route Selector (RS)
- Navigator

References to stimulus/response sequence throughout this SRS are structured accordingly to the following guidelines:

- [OPT]: Stimulus for optional recommendations.
- AND: Both stimulus before and after [AND] must be both true.
- OR: Either the stimulus before or after [OR] is true, or both stimulus before and after [OR] are true.

3.2.1. Base Map Displayer

The Base Map Displayer (BMD) is responsible for displaying the base map. The subsystem is located in the SPA and acts as the base for other overlays. BMD contains two modes ”Default” and ”Support”.

Stimulus/Response sequence

Stimulus	Response
Idle	Displays the base tile map.
Move Map	Change map position and show new map display area
Zoom In/Out Map	Change map scale and show new map display area
Button ”Enter Start Location and Destination” is pressed	Send button press information to RS and change mode to ”Support” mode.

Navigation ends instance OR RS ends instance	Change mode to ”Default” mode.
”Navigation Completed” message from Navigator [OPT]	Display relevant message to notify end of navigation

Table 2: Stimulus/Response of BMD

Associated Functional requirements

Requirement ID	Description
REQ-BMD-1	The BMD shall be initialised on SPA startup. The BMD shall be in ”Default” mode when initialised.
REQ-BMD-2	The BMD shall always be active during SPA runtime.
REQ-BMD-3	The BMD shall act as the base for other visual overlays.
REQ-BMD-4	If the only active interface The BMD shall act as the base for other visual overlays.
OPT-BMD-5	The BMD should display a message when navigation is completed.

Table 3: Functional Requirements of BMD

3.2.2. Shade Overlay

The Shade Overlay is responsible for displaying the shade visualisations. The subsystem is located in the SPA and provides shade visualisations by either through a toggle-able button and a slider.

Stimulus/Response sequence

Stimulus	Response
Toggled On	Displays shade visualisations.
Toggled Off	Turns off current shade visualisations if it exists.
Slider is moved	Displays the shade visualisations with the chosen time.
BMD changes map display area	Display relevant shade visualisations based on new map display area
Subsystem requests for shade information [OPT]	Send shade information to the subsystem
Change in STATE [OPT]	Toggled On or Toggle Off based on previous Shade Overlay toggle.

Table 4: Stimulus/Response of SO

Associated Functional requirements

Requirement ID	Description
REQ-SO-1	The Shade Overlay shall display shade visualisations in the SPA
REQ-SO-2	The Shade Overlay shall receive current day shade information from the GDBS on SPA start up.
REQ-SO-3	The Shade Overlay shall display shade visualisations depending on the current mode.
REQ-SO-3.1	If The BMD is in "Default" mode, The Shade Overlay shall display the "Button and Shade Slider" interface.
REQ-SO-3.2	Then, If the RS is in "Input" mode, the Shade Overlay shall be toggled off.
REQ-SO-3.3	Else, The Shade Overlay shall display the "Button" interface.
OPT-SO-4	The Shade Overlay shall provide shade information to other subsystems when prompted.

Table 5: Functional Requirements of SO

3.2.3. Route Selector

The Route Selector is responsible for receiving the needed inputs to calculate and display the calculated routes. This enables the user to choose and select one route before the start "Navigation" mode. The subsystem is located in the SPA. The subsystem contains two modes: "Input" mode and "Selection" mode.

Stimulus/Response sequence

Stimulus	Response
Button "Enter Start Location and Destination" is pressed OR Button "Go Back" is pressed during "Selection" mode	Change mode to "Input" mode and only enable "Input" mode features
Button "Calculate Routes" is pressed	Change mode to "Selection" mode and only enable "Selection" mode features
Button "Start Navigation" is pressed	End RS instance.

Table 6: Stimulus/Response of RS

Stimulus	Response
Idle	Displays 'User Input' interface.
Either Start Location or Destination button is pressed	Select and show the relevant pin and allow modification of the location

Start Location confirmed	Select Destination Location Pin
Destination confirmed	Display the "Calculate Routes" Button
Button "Calcualte Routes" is pressed	Send input and time data to the GDBS
Button "Go Back" is pressed	End RS instance

Table 7: Stimulus/Response of "Input" mode in RS

Stimulus	**Response**
Waiting for all routes from GDBS	Displays 'Calculating Routes' interface.
All routes received from GDBS	Displays 'Route Selection' interface.
Route Selection	Displays all information for the selected route
Button "Start Navigation" is pressed	Send the selected route data to the Navigator
Button "Go Back" is pressed	Enter both previous start and location data for "input" mode

Table 8: Stimulus/Response of "Selection" mode in RS

Associated Functional requirements

Requirement ID	**Description**
REQ-RS-1	The RC shall receive user start location and destination data during "Input" mode.
REQ-RS-2	The RC shall send all needed data to the GDBS. (for calculating routes using a path finding algorithm)
REQ-RS-3	The RC shall receive three routes from the GDBS.
REQ-RS-3.1	The RC shall receive a route with the shortest path.
REQ-RS-3.2	The RC shall receive a route with the highest shadow coverage.
REQ-RS-3.3	The RC shall receive a route with the least cost. (A route with both "cost-efficient" time and shade coverage)
REQ-RS-4	The RC shall display all routes it receives back from the GDBS during "Selection" mode.
REQ-RS-5	The RC shall provide the selected route to the Navigator system when the "Start Navigation" Button is pressed.

Table 9: Functional Requirements of RS

3.2.4. Navigator

The Navigator subsystem is responsible for providing navigation services for the system. The subsystem is located in the SPA and provides route visualization and navigation support. Navigator is in ”Navigation” mode while it is active.

Stimulus/Response sequence

Stimulus	Response
Route is received from RS	Displays the ”Navigation” Interface with the route overlay.
Changes in Current User Location	Change user icon Position on the map, based on current location.
Changes in Current Time AND User location has not changed	Change ETA of navigation.
Current User Location is within 100m of Destination	End Navigation instance
Timeout Triggered	End Navigation instance.
Wandering away from selected route detected [OPT]	If ”Lateness” time exceeds ”Route Relevancy Interval” time, send current location and destination to GDBS (for recalculation)
Updated route received from GDBS [OPT]	Apply changes to the route overlay.

Table 10: Stimulus/Response of Navigator

Associated Functional requirements

Requirement ID	Description
REQ-NAV-1	The Navigator provide navigation services of the SPA
REQ-NAV-2	The Navigator shall start navigation mode when it receives route data
REQ-NAV-3	The Navigator shall change navigation contents based on stimulus mentioned in Table 10

Table 11: Functional Requirements of Navigator

3.3. Performance Requirements

This subsection specifies both the static and the dynamic numerical requirements placed on the system or on human interaction with the software as a whole.

Requirement ID	Description
REQ-PR-1	Map rendering shall complete within 2 seconds under normal network conditions.

REQ-PR-2	Route calculation shall be completed within 3 seconds.
REQ-PR-3	The Shadow Overlay slider must update within 1 seconds when adjusted
REQ-PR-4	The system shall support concurrent access by multiple users.

Table 12: Performance Requirements

3.4. Design Constraints

This section describes design constraints that can be imposed by other standards, hardware limitations, etc.

Requirement ID	Description
REQ-DC-1	The system shall operate within standard web browser environments.
REQ-DC-2	The system shall operate within the predefined geographic region.
REQ-DC-3	The system shall not require specialized hardware beyond consumer-grade devices.
REQ-DC-4	The system shall rely on external cloud cover data sources.

Table 13: Design Constraints

3.5. Software System Attributes

This subsection describes each attribute of software that serves as a requirement. Factors required to guarantee a defined reliability, availability and maintainability of the system at time of delivery.

3.5.1. Reliability

Requirement ID	Description
REQ-SSR-1	The system shall reliably show correct map visualisations 90% of the time.
REQ-SSR-2	The system shall reliably show correct route calculations 90% of the time.
REQ-SSR-3	The system shall detect and report subsystem errors 95% of the time.

Table 14: Reliability Requirements

3.5.2. Availability

Requirement ID	Description
REQ-SSA-1	The system shall operate 24 hours during the day.

REQ-SSA-2	The SPA shall remain accessible whenever the GDBS is operational.

Table 15: Availability Requirements

3.5.3. Maintainability

Requirement ID	Description
REQ-SSM-1	The system shall use modular subsystem architecture.
REQ-SSM-2	Test environments shall be built to test the system.

Table 16: Maintainability Requirements